

RAM CHARAN VANGA

RA2511027010164

Sub ject : DSA

Unit 3, S10, SLO2 - Implementation of Queue using Linked List

Build the code for Implementation of Queues using Linked List (Pointer Implementation):

a. Define the structure for a node

b. Write a function for enqueue operation with arguments struct queue *q and int val

c. Write a function for dequeue operation with arguments struct queue *q

Answer: /* a. Node and Queue Structures */

```
struct node {  
    int data;  
    struct node *next;  
};
```

```
struct queue {  
    struct node *front;  
    struct node *rear;  
};
```

/* b. Enqueue Function */

```
void enqueue(struct queue *q, int val) {  
    struct node *ptr = (struct node *)malloc(sizeof(struct node));  
    if (ptr == NULL) {  
        printf("Memory Allocation Failed!\n");  
        return;  
    }  
    ptr->data = val;  
    ptr->next = NULL;  
    if (q->front == NULL) {  
        q->front = q->rear = ptr;  
    } else {  
        q->rear->next = ptr;  
        q->rear = ptr;  
    }  
    printf("Enqueued to linked queue: %d\n", val);  
}
```

/* c. Dequeue Function */

```
int dequeue(struct queue *q) {  
    if (q->front == NULL) {  
        printf("Queue Underflow! Linked queue is empty\n");  
        return -1;  
    }  
    struct node *ptr = q->front;
```

```
int val = ptr->data;
q->front = q->front->next;
if (q->front == NULL) {
    q->rear = NULL;
}
free(ptr);
return val;
}
```